Weight Space Representation Learning via Neural Field Adaptation

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Take-Aways

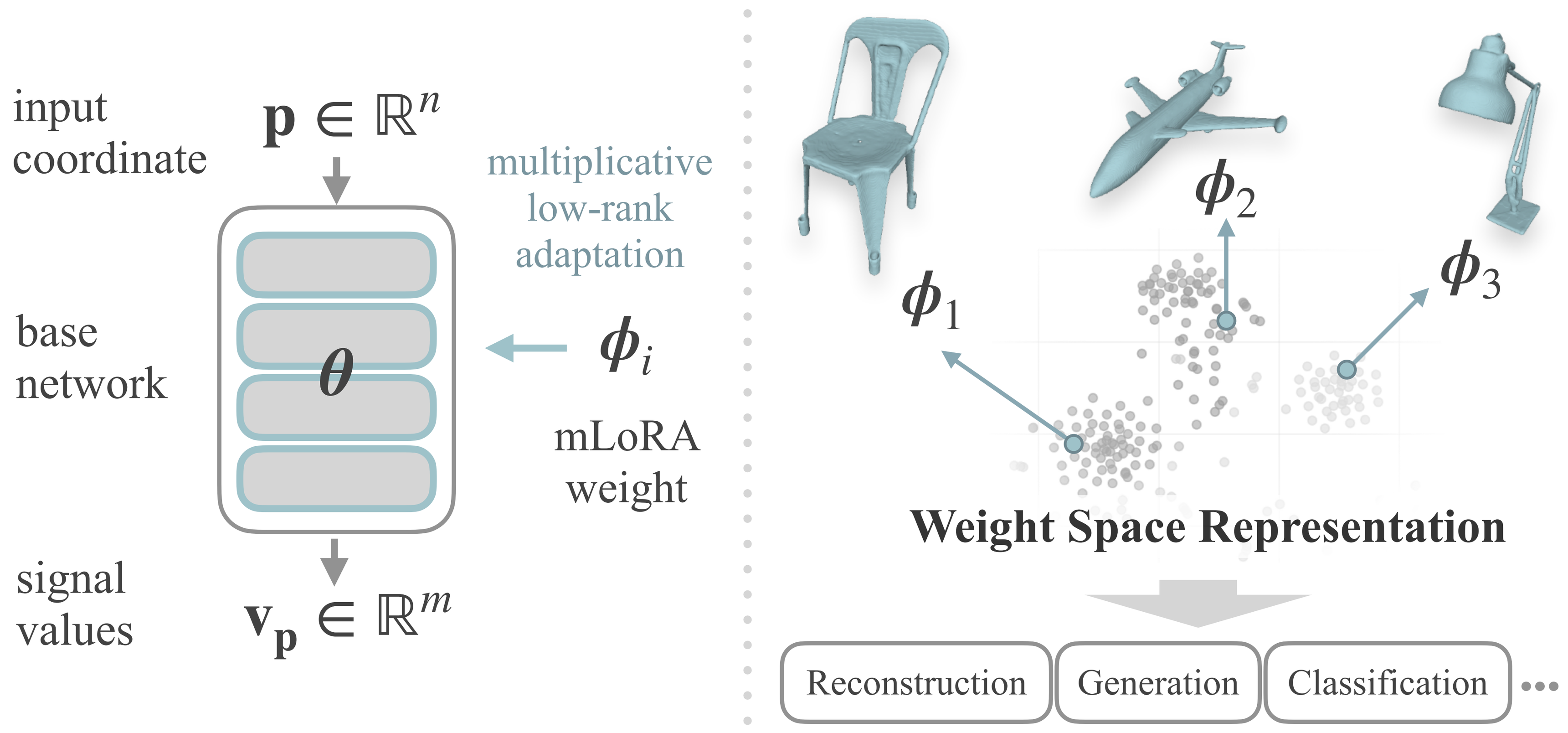
1 **Symmetry-broken** weights exhibit **semantic structure** and serve as **effective data representations**;

2 **Structured** weight space enables **effective weight-space generation**.

Project Page **Zoom**

1 Can Weights Serve as Representations?

▪ **A fitted neural field already encodes its data.** An image or shape lives entirely in its parameters.

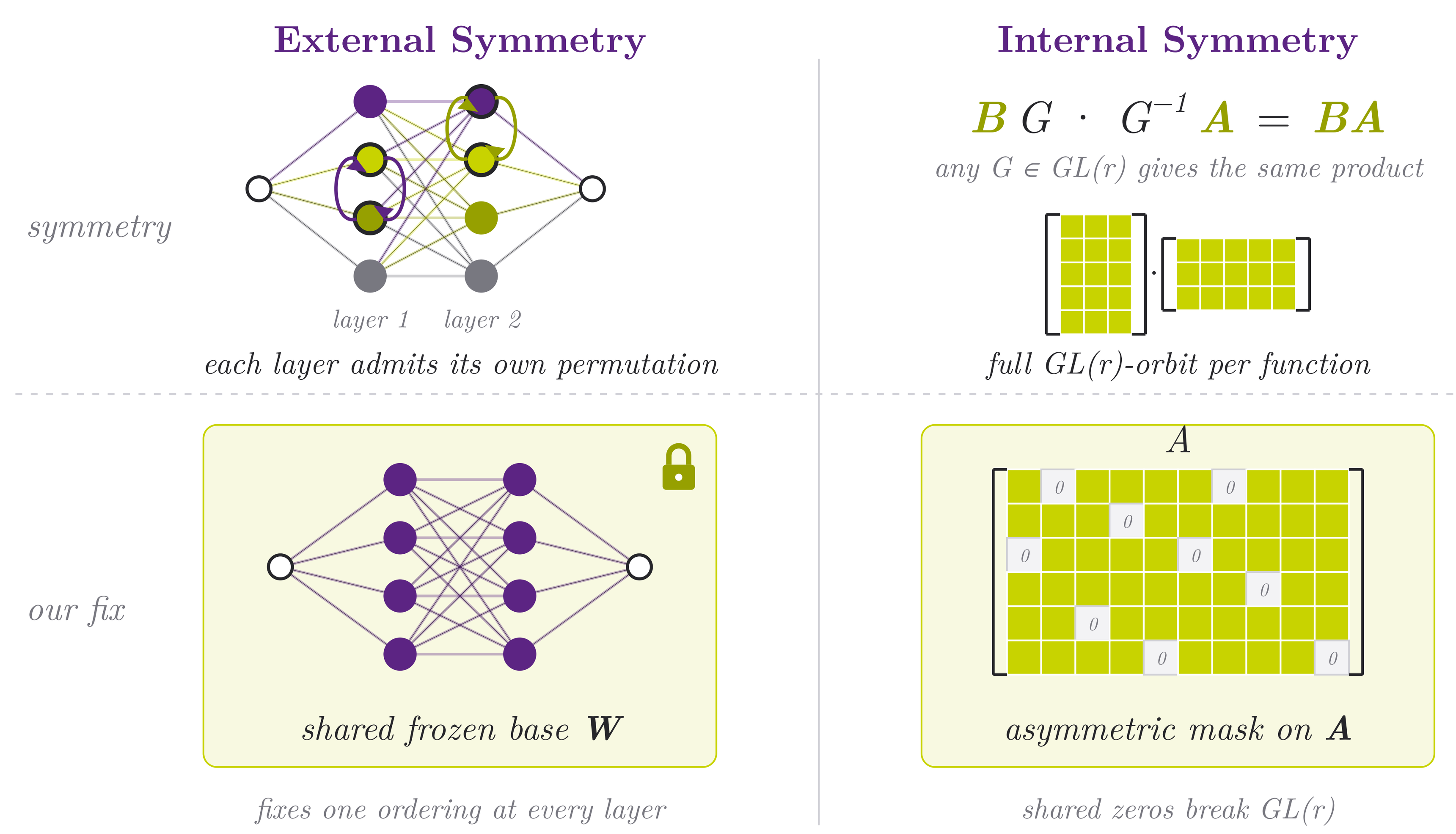


▪ **Representation Paradigm** A frozen base neural field is adapted by per-instance mLoRA weights ϕ_i; the weights *are* the representation.

▪ **Multiplicative LoRA** (mLoRA) instead of additive $\mathbf{W}' = \mathbf{W} \odot \mathbf{BA}$ (element-wise multiplication).

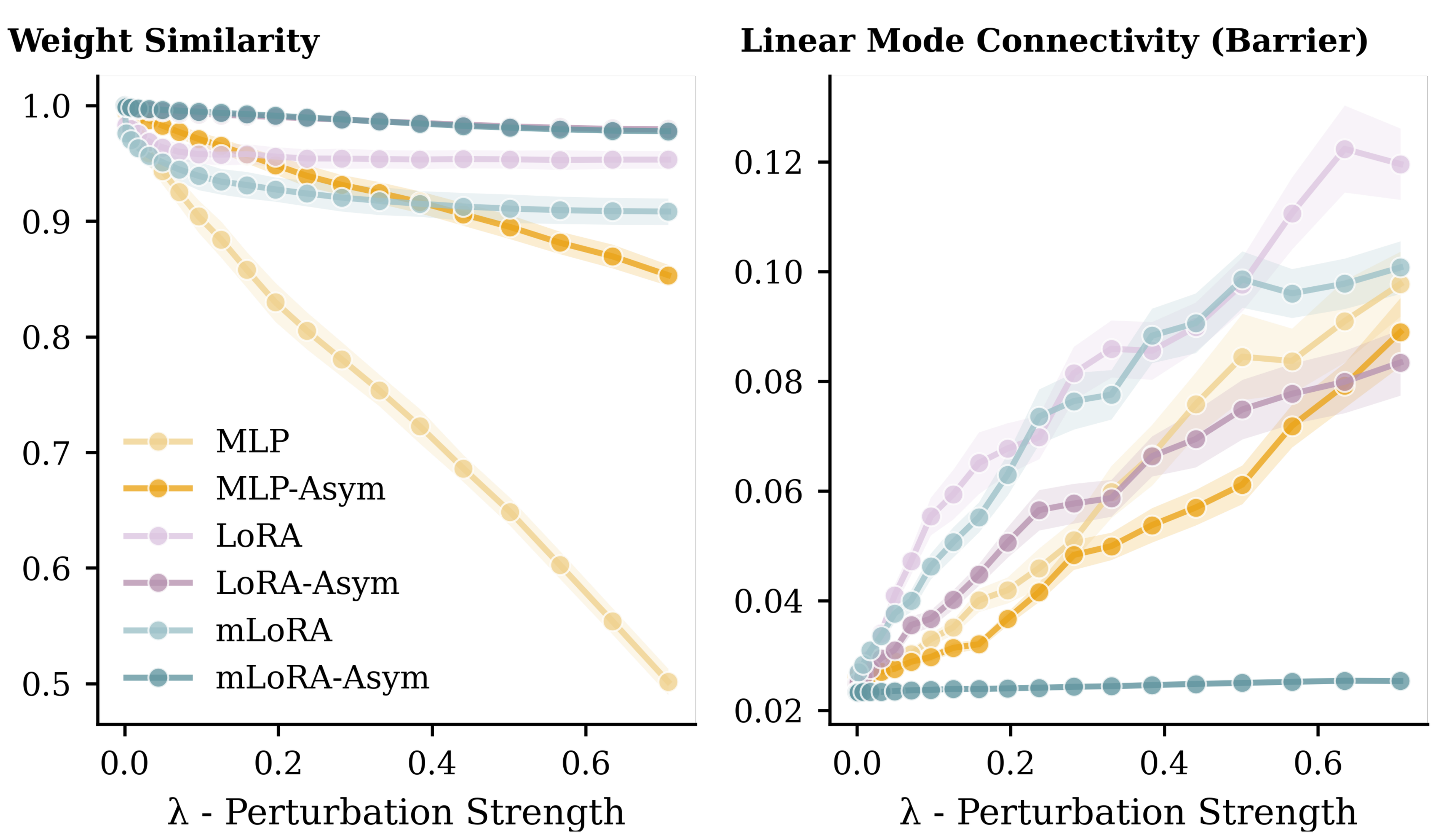
3 Breaking Weight Space Symmetries

▪ We constrain the neural field optimization to **break symmetries**.

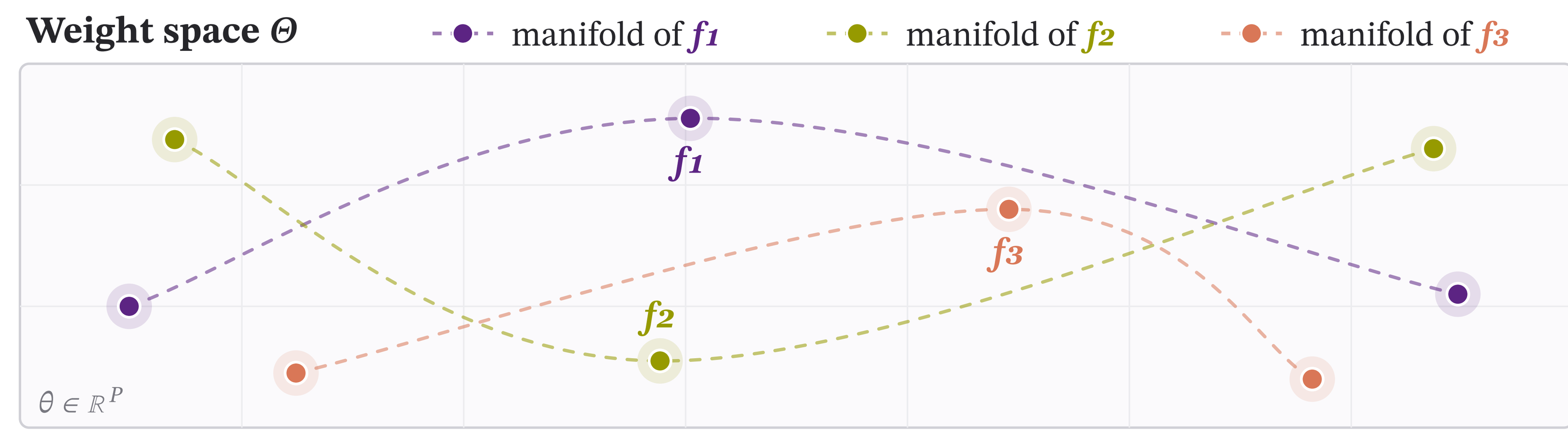


4 A Structured, Learnable Geometry

▪ Across different initializations, mLoRA-Asym fits converge to the **same linear mode**: high weight similarity, near-zero loss barrier. This smooth geometry is what makes the weight space **learnable**.



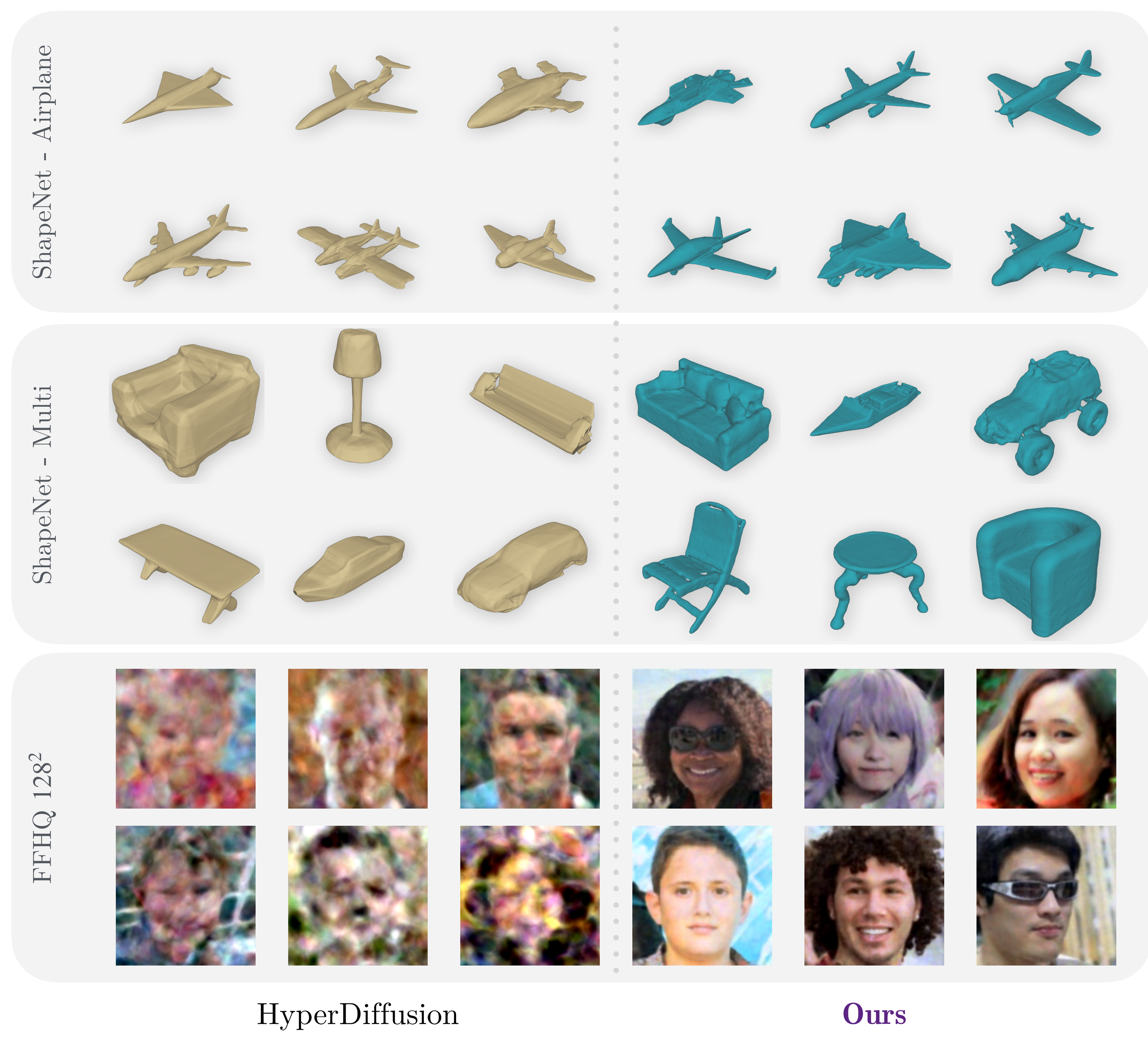
2 Challenge: Weight Space Symmetries



▪ **Symmetries complicate weight space structure. Functionally identical** networks could land **arbitrarily far** apart in weight space. The distribution becomes wildly **multi-modal**.

5 Generating New Neural Fields

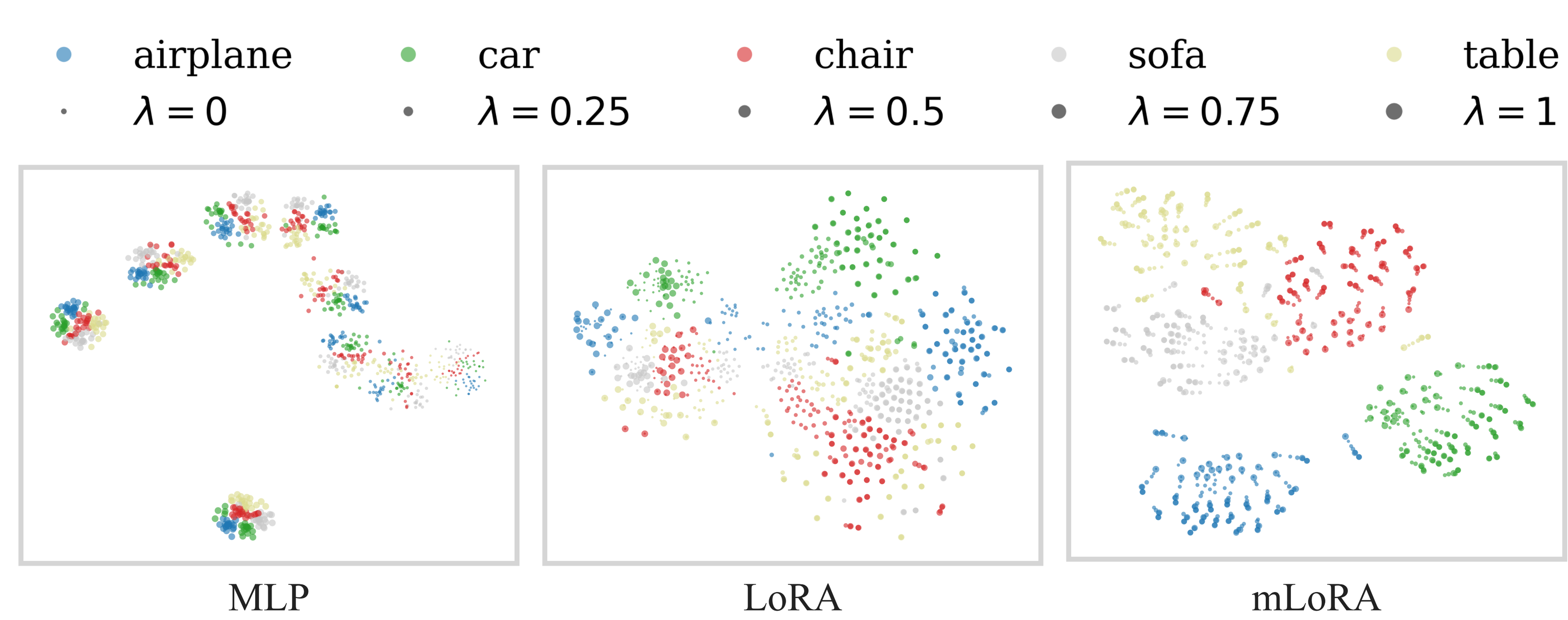
▪ A **diffusion model** trained directly on the mLoRA weight set samples new weights, and therefore new neural fields.



First successful weight-space generation of images at 128×128.

6 Weights Encode Semantics

▪ t-SNE visualization of weight spaces under different initializations. (color=category, size=initialization)



▪ For MLP the geometry is dominated by initialization (initialization → category → instance), whereas for mLoRA the semantic factor dominates (category → instance → initialization).